



ರೈ ತಾಂತ್ರಿಕ ವಿಶ್ವವಿದ್ಯಾಲಯ
Rai Technology University
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DEEP LEARNING

Assignment No: 2

MNIST Assignment

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MNIST - Modified National Institute of Standards and Technology database.

Report :-

Introduction.

The goal of this assignment is to build a neural network using TensorFlow / Keras to classify handwritten digit images (0-9) from the MNIST dataset. MNIST is a well-known dataset containing 60,000 training images and 10,000 test images, each a 28×28 pixel grayscale image of a handwritten digit along with its correct label.

Data set Exploration.

The dataset is loaded directly using `keras.datasets.mnist.load_data()`.

Sample images were displayed along with their labels (5, 0, 4, 19) to confirm the data loaded correctly. Each image is a grid of pixel values ranging from black to white. Before training, all pixel values

were normalized to a range of 0 to 1 by 255, which helps the neural network train more efficiently.

Model Design.

A simple feed-forward neural network (sequential model) was built with the following architecture:

- Flatten layer: converts each 28×28 image into a single 784-value vector.
- Dense hidden layer: 128 neurons with ReLU activation, used to learn patterns in the pixel data.
- Dense output layer: 10 neurons with ReLU activation, used to learn patterns in the pixel data.
- Dense output layer: 10 neurons with ReLU

The model is sparse categorical cross-entropy loss function, which is standard for multi-class classification problems.

It is trained for 5 epochs with a validation split of 20%.

Model Evaluation.

Training accuracy improved steadily across the 5 epochs, rising from 91.70% (Epoch 1) to 98.54% (Epoch 5), while training loss dropped from 0.2858 to 0.0478. After training, the model was evaluated on the 10,000 - unseen test images & achieved a test accuracy of 97.43%, meaning it correctly classified 97.43% of digits it had never seen before showing strong generalization.

Training Visualization.

The training and validation accuracy & loss were plotted across epochs. Training accuracy rose steadily throughout, while validation accuracy rose quickly at first & then flattened out around epoch 3, staying close to the training curve. Similarly, both training & validation loss dropped sharply in the first epoch & then leveled off, with validation loss flattening near 0.09-0.10 -

- while training loss kept decreasing slightly. the small, growing gap between training and validation curves suggests the model was beginning to slightly overfit by the later epochs.

Testing on sample Images.

Five random images from the test set were passed through the trained model to compare actual labels with predicted labels.

The results are:-

Actual	predicted	correct?
7	7	✓
2	2	✓
1	1	✓
0	0	✓
4	4	✓

All 5 out of 5 images were correctly classified, demonstrating the model's paraternal reliability at recognizing handwritten digits it had not seen during training.

P, T, D

Experiment: Adding Dropout.

As the required Experiment, a Dropout layer (rate = 0.3) was added to the network to reduce overfitting by randomly disabling neurons during training. The baseline model achieved 97.43% test accuracy, while the model with dropout achieved 97.49% - a slight improvement. This shows dropout helped the model generalize a bit better, though the gain is small since MNIST is a simple dataset & the baseline was already performing well.

Conclusion.

The neural network classified MNIST digits with 97.43% test accuracy, correctly predicting all 5 sample images. Adding dropout slightly improved accuracy to 97.49% showing better generalization. The project covered the full deep learning workflow from data loading to model.

Evaluation.